Percentage Increase and Decrease

Name _____ Date _____ Calculator allowed Version 1 Scan to practise this topic online

KEY WORDS percentage *(ōrau)* · multiplier *(tau whakarea)* · increase *(pikinga)* · decrease *(hekenga)*

A Percentage of an amount ●○○

1. Work out:

a) 10% of 350 = _____ **b)** 25% of 84 = _____ **c)** 5% of 260 = _____ **d)** 15% of 80 = _____

e) 40% of $65 = _____ **f)** 12% of 450 = _____ **g)** 7.5% of 240 = _____ **h)** 120% of 35 = _____

B Multipliers ●○○

Remember: To increase by 15%, multiply by 1.15. To decrease by 15%, multiply by 0.85.

2. Write the single multiplier for each change.

a) Increase by 20% _____ **b)** Decrease by 20% _____ **c)** Increase by 5% _____

d) Decrease by 35% _____ **e)** Increase by 12.5% _____ **f)** Decrease by 2% _____

g) Increase by 150% _____ **h)** Decrease by 0.5% _____ **i)** Increase by 7.5% _____

3. Describe the change each multiplier makes. For example, × 1.3 is an increase of 30%.

a) × 1.4 _____ **b)** × 0.75 _____ **c)** × 1.08 _____

d) × 0.4 _____ **e)** × 1.015 _____ **f)** × 0.925 _____

C Increasing and decreasing an amount ●●○

4. Increase:

a) 60 by 10% **b)** 250 by 30% **c)** $48 by 25%

d) 720 by 5% **e)** $36 by 15% **f)** 1,400 by 2.5%

5. Decrease:

a) 90 by 10% **b)** 400 by 35% **c)** $64 by 25%

d) 850 by 8% **e)** $129 by 40% **f)** 2,600 by 1.5%

D Percentage change ●●○

Remember: Percentage change = change ÷ original amount × 100

6. Find the percentage change. Say whether it is an increase or a decrease.

a) from 40 to 50 **b)** from 80 to 68 **c)** from $250 to $290 **d)** from 64 to 48

e) from 1,200 to 1,290 **f)** from $45 to $36 **g)** from 160 to 400 **h)** from 72 to 63

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

E Finding the original amount ●●●

⚠ **Watch out:** The percentage was taken from the original amount, not from the new amount.

7. Find the original amount.

a) After a 10% increase, the amount is 55.

b) After a 20% decrease, the amount is 64.

c) After a 25% increase, the price is $90.

d) After a 30% discount, the price is $84.

e) After a 6% increase, the amount is 1,272.

f) After a 15% decrease, the mass is 34 kg.

g) With 15% GST added, the price is $69.

h) After a 2.5% pay rise, the wage is $28.70.

F Problems ●●●

8. A jacket costs $180. In a sale it is reduced by 35%. Find the sale price.

9. Mere earns $26 per hour. She gets a 4% pay rise. Find her new hourly rate.

10. A plumber charges $640 plus 15% GST. Find the total bill.

11. A school roll grows from 850 students to 918 students. Find the percentage increase.

12. A phone bought for $1,200 is sold a year later for $780. Find the percentage decrease in its value.

13. A shop buys a jacket for $60 and sells it for $84. Find the percentage profit.

14. A TV costs $1,020 in a sale, after a 15% discount. What was the price before the sale?

15. A café bill is $92, including 15% GST. How much of the bill is GST?

16. Tāne's rent went up by 8% to $486 a week. How much more does he now pay each week?

17. A town has 24,000 people. Its population grows by 5% each year. Find the population after 2 years.

★ Extension

18. A shop raises a price of $80 by 20%. Later it reduces the new price by 20%.

a) Find the final price.

b) Explain why the final price is not $80.

19. A price rises by 25%. By what percentage must the new price fall to get back to the original price?

20. A car is worth $32,000. It loses 15% of its value each year. Find its value after 3 years.

21. The length of a rectangle is increased by 10% and its width is decreased by 10%. Find the percentage change in its area.

22. Two shops sell the same $400 bike. Shop A takes 25% off. Shop B takes 15% off, then another 10% off the new price. Which shop is cheaper, and by how much?

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

Percentage Increase and Decrease

Name _____ Date _____ Calculator allowed Version 2

Scan to practise this topic online

KEY WORDS percentage *(ōrau)* · multiplier *(tau whakarea)* · increase *(pikinga)* · decrease *(hekenga)*

A Percentage of an amount ●○○

1. Work out:

a) 10% of 470 = _____ **b)** 25% of 96 = _____ **c)** 5% of 340 = _____ **d)** 15% of 60 = _____

e) 40% of $95 = _____ **f)** 12% of 350 = _____ **g)** 7.5% of 320 = _____ **h)** 140% of 45 = _____

B Multipliers ●○○

Remember: To increase by 15%, multiply by 1.15. To decrease by 15%, multiply by 0.85.

2. Write the single multiplier for each change.

a) Increase by 40% _____ **b)** Decrease by 40% _____ **c)** Increase by 8% _____

d) Decrease by 45% _____ **e)** Increase by 17.5% _____ **f)** Decrease by 4% _____

g) Increase by 120% _____ **h)** Decrease by 1.5% _____ **i)** Increase by 6.5% _____

3. Describe the change each multiplier makes. For example, × 1.3 is an increase of 30%.

a) × 1.6 _____ **b)** × 0.85 _____ **c)** × 1.04 _____

d) × 0.3 _____ **e)** × 1.025 _____ **f)** × 0.875 _____

C Increasing and decreasing an amount ●●○

4. Increase:

a) 80 by 10% **b)** 350 by 20% **c)** $56 by 25%

d) 640 by 5% **e)** $44 by 15% **f)** 1,800 by 2.5%

5. Decrease:

a) 70 by 10% **b)** 600 by 45% **c)** $84 by 25%

d) 950 by 6% **e)** $135 by 40% **f)** 3,400 by 1.5%

D Percentage change ●●○

Remember: Percentage change = change ÷ original amount × 100

6. Find the percentage change. Say whether it is an increase or a decrease.

a) from 50 to 60 **b)** from 60 to 51 **c)** from $350 to $392 **d)** from 96 to 72

e) from 1,600 to 1,720 **f)** from $65 to $52 **g)** from 140 to 420 **h)** from 56 to 49

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

E Finding the original amount ●●●

⚠ **Watch out:** The percentage was taken from the original amount, not from the new amount.

7. Find the original amount.

a) After a 10% increase, the amount is 77.

b) After a 20% decrease, the amount is 96.

c) After a 25% increase, the price is $60.

d) After a 30% discount, the price is $63.

e) After a 4% increase, the amount is 1,560.

f) After a 15% decrease, the mass is 51 kg.

g) With 15% GST added, the price is $92.

h) After a 2.5% pay rise, the wage is $32.80.

F Problems ●●●

8. A jacket costs $240. In a sale it is reduced by 35%. Find the sale price.

9. Sione earns $28 per hour. He gets a 3% pay rise. Find his new hourly rate.

10. An electrician charges $520 plus 15% GST. Find the total bill.

11. A school roll grows from 750 students to 795 students. Find the percentage increase.

12. A laptop bought for $1,600 is sold a year later for $1,120. Find the percentage decrease in its value.

13. A shop buys a pair of shoes for $80 and sells them for $108. Find the percentage profit.

14. A fridge costs $1,275 in a sale, after a 15% discount. What was the price before the sale?

15. A restaurant bill is $138, including 15% GST. How much of the bill is GST?

16. Aroha's rent went up by 5% to $504 a week. How much more does she now pay each week?

17. A town has 15,000 people. Its population grows by 6% each year. Find the population after 2 years.

★ Extension

18. A shop raises a price of $60 by 10%. Later it reduces the new price by 10%.

a) Find the final price.

b) Explain why the final price is not $60.

19. A price falls by 20%. By what percentage must the new price rise to get back to the original price?

20. A car is worth $25,000. It loses 20% of its value each year. Find its value after 3 years.

21. The length of a rectangle is increased by 20% and its width is decreased by 20%. Find the percentage change in its area.

22. Two shops sell the same $600 TV. Shop A takes 30% off. Shop B takes 20% off, then another 10% off the new price. Which shop is cheaper, and by how much?

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.